Hilbert Spaces

H.1

Solution:

□

H.2

Solution:

□

H.3

Solution:

□

H.4

Solution:

□

H.5a

Solution:

□

H.11

Solution:

□

H.12

Solution:

□

H.13

Solution:

□

H.15

$$(u \mid v) = \int_{-\infty}^{\infty} u(x)v(x)\frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x$$

Determine the first three Hermite polynomials

Solution:

Using Grahm-Schmidt with $u_n = [1, x, x^2]$

$$\varphi_0 = 1$$

$$(\varphi_0 \mid \varphi_0) = \int_{-\infty}^{\infty} \frac{e^{-x^2}}{\sqrt{\pi}} \,\mathrm{d}x = 1$$

$$\psi_0 = \frac{\varphi_0}{(\varphi_0 \mid \varphi_0)} = 1$$

Next

$$\varphi_1 = u_1 - P_{[\varphi_0]} u_1 = u_1 - \frac{(\varphi_0 \mid u_1)}{(\varphi_0 \mid \varphi_0)} \varphi_0 = x - \frac{(1 \mid x)}{(1 \mid 1)} 1 =$$

$$x - \underbrace{\int_{-\infty}^{\infty} x \frac{e^{-x^2}}{\sqrt{\pi}} \,\mathrm{d}x}_{=0 \text{ (odd function)}} = x$$

$$\psi_1 = \frac{\varphi_1}{(\varphi_1 \mid \varphi_1)} = \frac{x}{(x \mid x)}$$

$$(x \mid x) = \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}} \,\mathrm{d}x = -\frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} (x) \left(-2x e^{-x^2}\right) =$$

$$= -\frac{1}{2\sqrt{\pi}} \left(\left[x e^{-x^2} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-x^2} \,\mathrm{d}x \right) = \frac{1}{2}$$

Finally this gives

$$\psi_1 = \frac{x}{(x \mid x)^{\frac{1}{2}}} = \sqrt{2} x$$

Next

$$\varphi_2 = u_2 - P_{[\varphi_0,\varphi_1]}u_2 = x^2 - \frac{(1 \mid x^2)}{(1 \mid 1)}1 - \frac{(x \mid x^2)}{(x \mid x)}x =$$

$$= x^2 - \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x - \underbrace{\int_{-\infty}^{\infty} x^3 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x}_{=0\text{ odd function}} = x^2 - \frac{1}{2}$$

$$\psi_2 = \frac{\varphi_2}{(\varphi_2 \mid \varphi_2)^{\frac{1}{2}}} = \left(x^2 - \frac{1}{2}\right)\left(\int_{-\infty}^{\infty}\left(x^2 - \frac{1}{2}\right)^2 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x\right)^{-\frac{1}{2}} =$$

$$= \left(x^2 - \frac{1}{2}\right)\left(\int_{-\infty}^{\infty}\left(x^4 - x^2 + \frac{1}{4}\right) \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x\right)^{-\frac{1}{2}}$$

$$\text{(i)} \int_{-\infty}^{\infty} \frac{1}{4}\frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x = \boxed{\frac{1}{4}}$$

$$\text{(ii)} \int_{-\infty}^{\infty} -x^2 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x = \boxed{-\frac{1}{2}}$$

$$\text{(iii)} \int_{-\infty}^{\infty} x^4 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x = -\frac{1}{2\sqrt{\pi}}\int_{-\infty}^{\infty} (x^3)\left(-2xe^{-x^2}\right)\mathrm{d}x =$$

$$= \underbrace{\left[x^2 e^{-x^2}\right]_{-\infty}^{\infty}}_{=0} - \underbrace{\int_{-\infty}^{\infty} 3x^2 e^{-x^2}\,\mathrm{d}x}_{=\frac{3\sqrt{\pi}}{2}} = \boxed{\frac{3}{4}}$$

So

$$\psi_2 = \left(x^2 - \frac{1}{2}\right)\left(\frac{1}{4} - \frac{1}{2} + \frac{3}{4}\right)^{-\frac{1}{2}} = \left(x^2 - \frac{1}{2}\right)\sqrt{2}$$

The three first ortonormal Hermite polynomials are

$$\psi_0 = \boxed{1}$$

$$\psi_1 = \boxed{\sqrt{2}x}$$

$$\psi_2 = \boxed{\sqrt{2}\left(x^2 - \frac{1}{2}\right)}$$

□

H.18

$$\int_0^\pi (x - a\sin(x) - b\sin(2x))^2 \, \mathrm{d}x$$

is minimized.

Solution:

$$v \in \mathcal{M} = [\sin(x), \sin(2x)]$$

$\sin(x)$ and $\sin(2x)$ are orthogonal and we can use the projection formula which says that the minimum is obtained when

$$v = P_{[\mathcal{M}]}u$$

$$v_1 = a\sin(x) = P_{[\sin(x)]}u = \frac{(\sin(x) \mid x)}{(\sin(x) \mid \sin(x))}\sin(x)$$

$$v_2 = b\sin(2x)$$

From this, a is calculated as

$$a = \frac{(\sin(x) \mid x)}{(\sin(x) \mid \sin(x))}$$

$$(\sin(x) \mid x) = \int_0^\pi x\sin(x)\,\mathrm{d}x = -[x\cos(x)]_0^\pi + \int_0^\pi \cos(x)\,\mathrm{d}x = \pi$$

$$(\sin(x) \mid \sin(x)) = \int_0^\pi \sin^2 x\,\mathrm{d}x = \int_0^\pi \frac{1-\cos(2x)}{2}\,\mathrm{d}x = \frac{\pi}{2}$$

$$\downarrow$$

$$\boxed{a = \frac{\pi}{\frac{\pi}{2}} = 2}$$

and b is calculated in the same way as

$$b = \frac{(\sin(2x) \mid x)}{(\sin(2x) \mid \sin(2x))}$$

$$(\sin(2x) \mid x) = \int_0^\pi x\sin(2x)\,\mathrm{d}x = -\left[\frac{x\cos(2x)}{2}\right]_0^\pi + \int_0^\pi \frac{\cos(2x)}{2}\,\mathrm{d}x = -\frac{\pi}{2}$$

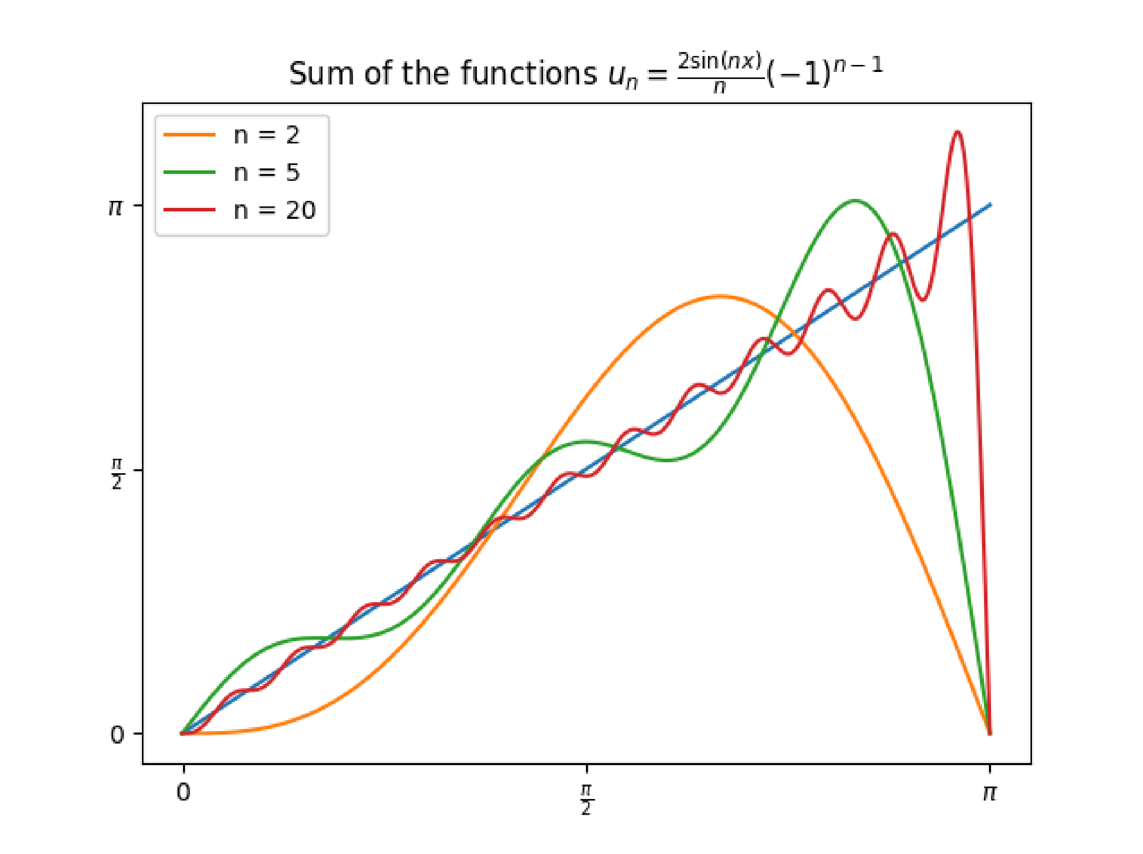
$$(\sin(2x) \mid \sin(2x)) = \int_0^\pi \sin^2(2x)\,\mathrm{d}x = \int_0^\pi \frac{1-\cos(2x)}{2}\,\mathrm{d}x = \frac{\pi}{2}$$

$$\downarrow$$

$$\boxed{b = -\frac{\frac{\pi}{2}}{\frac{\pi}{2}} = -1}$$

In general, the functions $v \in \mathcal{M} = \{\sin(nx)\}_{n=1}^{\infty}$ that best approximate $u(x) = x$ at the interval $[0, \pi]$ is given by

$$u_n = \frac{2(-1)^{n-1} \sin nx}{n}$$



□

H.19

(a) Give an orthogonal base in $\mathcal{V}$ with the scalar product

$$(u \mid v) = \int_{-1}^{1} u(x)v(x)\,\mathrm{d}x$$

(b) Let $u(x) = x^2 - x$. Determine $v \in \mathcal{V}$ so that

$$\int_{-1}^{1} |u(x) - v(x)|^2\,\mathrm{d}x$$

is minimized.

(c) Determine the lowest possible value of the integral in (b).

Solution:

(a) $\square$

H.20

Solution:

For this setup, the scalar product is defined as

$$(u \mid v) = \int_0^1 u(x)v(x)w(x)\,\mathrm{d}x = \int_0^1 u(x)v(x)x\,\mathrm{d}x$$

Find an orthogonal base

$$\varphi_0 = 1$$

$$(\varphi_0 \mid \varphi_0) = \int_0^1 x\,\mathrm{d}x = \frac{1}{2}$$

$$\boxed{\psi_0 = \frac{\varphi_0}{(\varphi_0 \mid \varphi_0)} = 2}$$

$$\varphi_1 = x - P_{[\varphi_0]}x = x - \frac{(\varphi_0 \mid x)}{(\varphi_0 \mid \varphi_0)}\varphi_0 = x - 2\int_0^1 x^2\,\mathrm{d}x = x - \frac{2}{3}$$

$$(\varphi_1 \mid \varphi_1) = \int_0^1 \left(x - \frac{2}{3}\right)^2 x\,\mathrm{d}x = \frac{1}{36}$$

$$\boxed{\psi_1 = \frac{\varphi_1}{(\varphi_1 \mid \varphi_1)} = 36x - 24}$$

Now $v \in \mathcal{M} = \{2, 36x - 24\}$

Are looking for

$$\int_0^1 \left(x^4 - a\psi_0 - b\psi_1\right)^2 x\,\mathrm{d}x$$

From which we can calculate

$$a\psi_0 = \frac{(\psi_0 \mid x^4)}{(\psi_0 \mid \psi_0)}\psi_0$$

$$a = \frac{(\psi_0 \mid u)}{(\psi_0 \mid \psi_0)}$$

$$(\psi_0 \mid u) = \int_0^1 2x^5 \,\mathrm{d}x = \frac{1}{3}$$

$$(\psi_0 \mid \psi_0) = \int_0^1 4x \,\mathrm{d}x = 2$$

$$\boxed{a = \frac{1}{6}}$$

and

$$b\psi_1 = \frac{(\psi_1 \mid x^4)}{(\psi_1 \mid \psi_1)}\psi_1$$

$$b = \frac{(\psi_1 \mid x^4)}{(\psi_1 \mid \psi_1)}$$

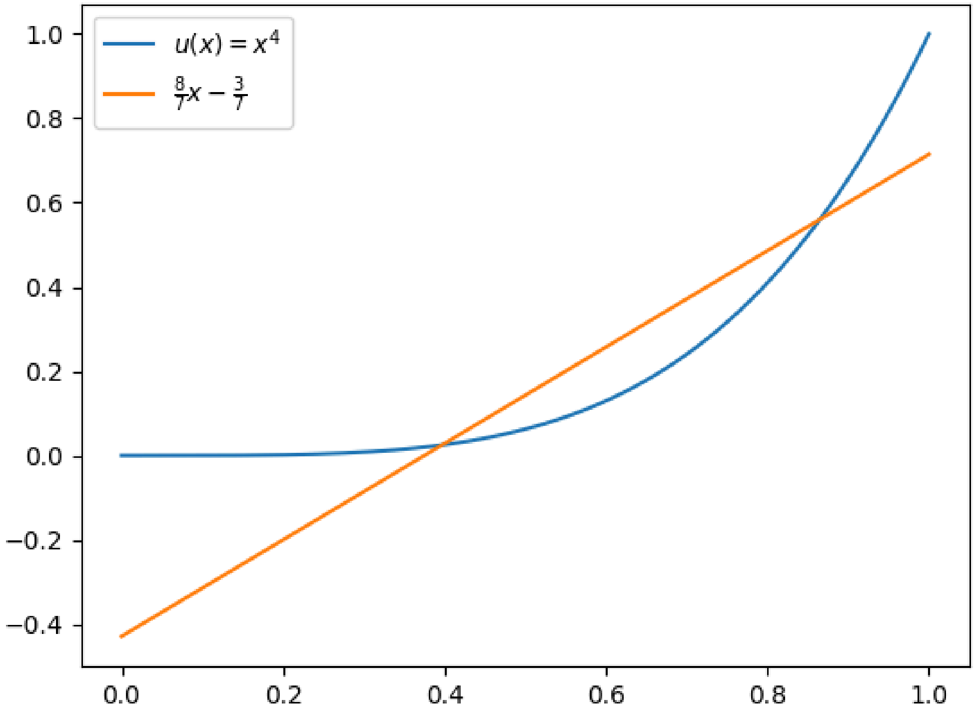
$$(\psi_1 \mid x^4) = \int_0^1 (36x^6 - 24x^5) \,\mathrm{d}x = \frac{36}{7} - \frac{24}{6} = \frac{8}{7}$$

$$(\psi_1 \mid \psi_1) = \int_0^1 (1296x^3 - 1728x^2 + 576x) = 36$$

$$\boxed{b = \frac{2}{63}}$$

The first degree polynomial that best approximates the function $u(x) = x^4$ in $I = [0, 1]$ is

$$\boxed{\frac{1}{6}\psi_0 + \frac{2}{63}\psi_1 = \frac{8}{7}x - \frac{3}{7}}$$



(b) We are now looking for a second degree polynomial which means our $\psi \in \mathcal{M} = \{1, x, x^2\}$ ψ_0 and ψ_1 were already calculated in (a) so

$$\varphi_2 = u_2 - P_{[\varphi_0,\varphi_1]}u_2 = x^2 - \frac{(\varphi_0 \mid x^2)}{(\varphi_0 \mid \varphi_0)}\varphi_0 - \frac{(\varphi_1 \mid x^2)}{(\varphi_1 \mid \varphi_1)}\varphi_1$$

$$\frac{(\varphi_0 \mid x^2)}{(\varphi_0 \mid \varphi_0)}\varphi_0 = 2(\varphi_0 \mid x^2) = 2\int_0^1 x^3 \, \mathrm{d}x = \frac{1}{2}$$

$$\frac{(\varphi_1 \mid x^2)}{(\varphi_1 \mid \varphi_1)}\varphi_1 = (\varphi_1 \mid x^2)(36x - 24)$$

Now calculating

$$(\varphi_1 \mid x^2) = \int_0^1 \left(x^4 - \frac{2x^3}{3}\right) \mathrm{d}x = \frac{1}{30}$$

which gives finally

$$\varphi_2 = x^2 - \frac{6}{5}x + \frac{3}{10}$$

$$(\varphi_2 \mid \varphi_2) = \int_0^1 \left(x^2 - \frac{6x}{5} + \frac{3}{10}\right)^2 x \,\mathrm{d}x = \frac{1}{600}$$

$$\boxed{\psi_2 = 600x^2 - 720x + 180}$$

We can now find the best approximation with

$$\int_0^1 \left(x^4 - a\psi_0 - b\psi_1 - c\psi_2\right)^2 x \,\mathrm{d}x$$

$$c\psi_2 = \frac{(\psi_2 \mid x^4)}{(\psi_2 \mid \psi_2)}\psi_2$$

$$c = \frac{(\psi_2 \mid x^4)}{(\psi_2 \mid \psi_2)}$$

$$(\psi_2 \mid x^4) = \int_0^1 (600x^2 - 720x + 180)x^4 x \,\mathrm{d}x = \frac{15}{7}$$

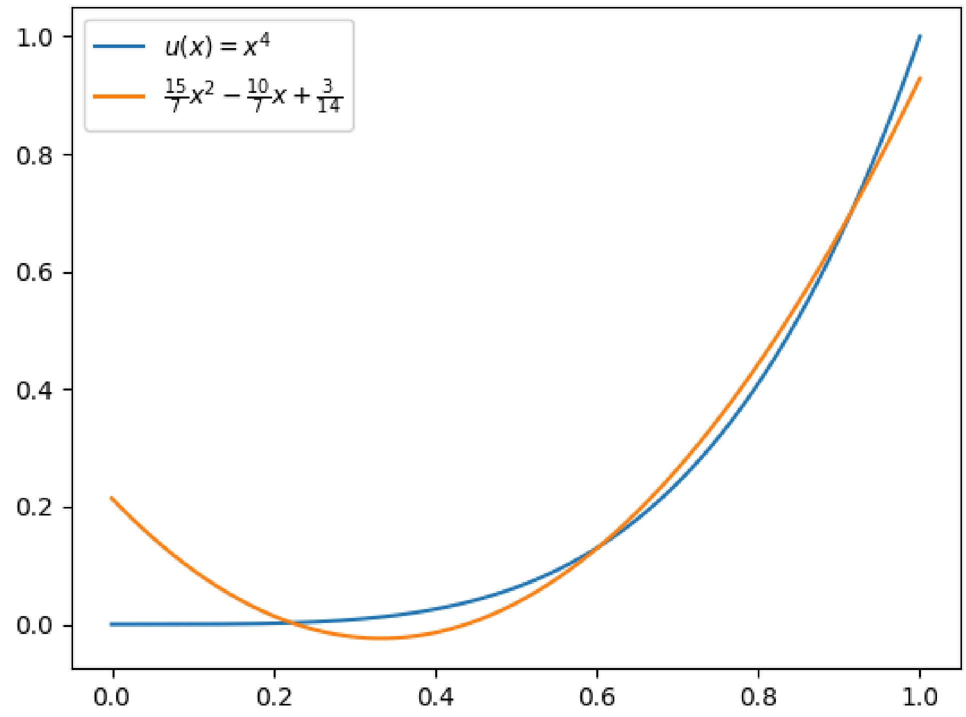
$$(\psi_2 \mid \psi_2) = \int_0^1 \left(600x^2 - 720x + 180\right)^2 x \,\mathrm{d}x = 600$$

$$\downarrow$$

$$c = \frac{15}{4200} = \frac{1}{280}$$

The best second degree polynomial to approximate $u(x) = x^4$ is then

$$\boxed{a\psi_0 + b\psi_1 + c\psi_2 = \frac{15}{7}x^2 - \frac{10}{7}x + \frac{3}{14}}$$



□

H.27

(a) Show that p_n is orthogonal to all polynomials of degree $p < n$

(b) Show that if p is a polynomial, not identically zero, such that $(p_k \mid p) = 0$ for $k = 0, 1, 2,$ then degree of $p \geq 3$.

Solution:

(a) All polynomials of degree m can be written

$$p = \sum_{k=0}^{m} c_k p_k$$

Gives

$$(p_n \mid p) = \left(p_n \mid \sum_{k=0}^{m} c_k p_k \right) = \sum_{k=0}^{m} c_k (p_n \mid p_k) = 0 \text{ for } m < n$$

since all polynomials are pairwise orthogonal

(b)

$$p = \sum_{k=0}^{m} c_k p_k$$

where

$$c_k = \frac{(p_k \mid p)}{(p_k \mid p_k)}$$

$$(p_k \mid p) = 0, \text{ for } k = 0, 1, 2 \to c_0 = c_1 = c_2 = 0$$

The restriction that $p \neq 0$ gives that $\deg p \geq 3$ □

H.30

Solution:

□

H.31

Solution:

□

H.33

$$f_n(x) = x^n, \quad x \in I = [0, 1], \quad n = 0, 1, ...$$

Determine $\|f_n\|_p$ for $p = 1, 2$ and ∞. Is it for any of these norms true that $f_n \to 0$ when $n \to \infty$? What is the pointwise limit function?

Solution:

$$\|f_n\|_1 = \int_0^1 |x^n| \, \mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1} \to 0, \quad n \to \infty$$

$$\|f_n\|_2 = \left(\int_0^1 |x^n|^2 \, \mathrm{d}x\right)^{\frac{1}{2}} = \left(\left[\frac{x^{2n+1}}{2n+1}\right]_0^1\right)^{\frac{1}{2}} = \frac{1}{\sqrt{2n+1}} \to 0, \quad n \to \infty$$

$$\|f_n\|_\infty = \sup_{x\in[0,1]} |f_n| = \sup_{x\in[0,1]} |x^n| = 1 \to 1, \quad n \to \infty$$

□

H.36

Solution:

□

H.38

Solution:

□

H.39

Solution:

□

H.45

Solution:

□

H.47

Solution:

□

H.50

$$\mathcal{A} = -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\right)$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(-1,1) \mid u \text{ and } u' \text{ bounded/limited}\}$$

(a) Determine a scalar product in which $\mathcal{A}$ is symmetric.

(b) Show that $\mathcal{A}$ is symmetric in the scalar product from (a), using the definition.

(c) Show that $1, x, 2x^2-1$ are eigenfunctions to $\mathcal{A}$ and determine the corresponding eigenvalues.

(d) Evaluate

$$\int_{-1}^{1} \frac{2x^2-1}{\sqrt{1-x^2}}\,\mathrm{d}x$$

Solution:

(a) The operator is a Sturm-Liouville operator

$$\mathcal{A}u = -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\right) = -\frac{1}{w}(pu')', \qquad w(x) = \frac{1}{\sqrt{1-x^2}}$$

and the scalar product is then

$$(u \mid v) = \int_{-1}^{1} \overline{u(x)}v(x)\frac{1}{w(x)}\,\mathrm{d}x$$

(b) Using the scalar product definition

$$(u \mid v) = \int \overline{u(x)} v(x) w(x) \, \mathrm{d}x, \qquad w(x) = \frac{1}{\sqrt{1-x^2}}$$

we get

$$(u \mid \mathcal{A}v) =$$

$$= \int_{-1}^{1} \overline{u(x)} [\mathcal{A}v(x)] \frac{1}{w(x)} \, \mathrm{d}x = \int_{-1}^{1} \overline{u(x)} \left[-\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} \left(\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} v(x) \right) \right] \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x =$$

$$= -\int_{-1}^{1} \overline{u(x)} \big(\sqrt{1-x^2} v'(x) \big)' \, \mathrm{d}x = -\underbrace{\Big[\overline{u(x)} \sqrt{1-x^2} v'(x) \Big]_{-1}^{1}}_{=0} + \int_{-1}^{1} \overline{u'(x)} \sqrt{1-x^2} v'(x) \, \mathrm{d}x =$$

$$= \int_{-1}^{1} \big(\overline{u'(x)} \sqrt{1-x^2} \big) v'(x) \, \mathrm{d}x = \underbrace{\Big[\big(\overline{u'(x)} \sqrt{1-x^2} \big) v(x) \Big]_{-1}^{1}}_{=0} - \int_{-1}^{1} \big(\overline{u'(x)} \sqrt{1-x^2} \big)' v(x) \, \mathrm{d}x =$$

$$= \int_{-1}^{1} -\big(\overline{u'(x)} \sqrt{1-x^2} \big)' v(x) \, \mathrm{d}x = \int_{-1}^{1} -\sqrt{1-x^2} \big(\overline{u'(x)} \sqrt{1-x^2} \big)' v(x) \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x =$$

$$= \int_{-1}^{1} \left[-\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} \left(\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} \overline{u(x)} \right) \right] v(x) \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x =$$

$$= \int_{-1}^{1} \Big[\mathcal{A} \overline{u(x)} \Big] v(x) \frac{1}{w(x)} \, \mathrm{d}x = \boxed{(\mathcal{A}u \mid v)}$$

(c)

$$\mathcal{A}u = \lambda u$$
$$u(x) = 1$$
$$\downarrow$$
$$\mathcal{A}\cdot 1 = \lambda_0 \cdot 1 \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\cdot 1\right) = \lambda_0 \cdot 1 \to 0 = \lambda_0 \cdot 1 \to \boxed{\lambda_0 = 0}$$

$$u(x) = x$$
$$\downarrow$$
$$\mathcal{A}x = \lambda_1 x \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}x\right) = \lambda_1 x \to -\sqrt{1-x^2}\frac{-2x}{2\sqrt{1-x^2}} = \lambda_1 x \to$$
$$\to x = \lambda_1 x \to \boxed{\lambda_1 = 1}$$

$$u(x) = 2x^2 - 1$$
$$\downarrow$$
$$\mathcal{A}(2x^2-1) = \lambda_2(2x^2-1) \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}(2x^2-1)\right) = \lambda_2(2x^2-1) \to$$
$$\to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(4x\sqrt{1-x^2}\right) = -\sqrt{1-x^2}\left(4\sqrt{1-x^2} - \frac{8x^2}{2\sqrt{1-x^2}}\right) = -4 + 4x^2 + 4x^2 =$$
$$= 8x^2 - 4 = \lambda_2(2x^2-1) \to 4(2x^2-1) = \lambda_2(2x^2-1) \to \boxed{\lambda_2 = 4}$$

(d) The integral can be rewritten with the scalar product in (a) as

$$\int_{-1}^{1}\frac{2x^2-1}{\sqrt{1-x^2}}\,\mathrm{d}x = (1 \mid 2x^2-1), \qquad w(x) = \frac{q}{\sqrt{1-x^2}}$$

Since the functions 1 and $2x^2-1$ are eigenfunctions, the scalar product between them are pairwise orthogonal and therefore $= 0$.

□

H.51

Solution:

□

H.52

Solution:

□

Operator

$$\mathcal{A} = -\ \texttt{laplace} = -\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$$

Solution:

$$(u \mid \mathcal{A}v) = \int_0^a \int_0^b \overline{u}(-v''_{xx} - v''_{yy}) \, \mathrm{d}x \, \mathrm{d}y =$$

$$(1) = -\int_0^b \underbrace{\left(\int_0^a \overline{u} v''_{xx} \, \mathrm{d}x \right)}_{\text{(i-x)}} \mathrm{d}y - \int_0^a \underbrace{\left(\int_0^b \overline{u} v''_{yy} \, \mathrm{d}y \right)}_{\text{(i-y)}} \mathrm{d}x$$

$$\text{(i-x)} = \underbrace{[\overline{u} v'_x]_0^a}_{=0} - \int_0^a \overline{u}'_x v'_x \, \mathrm{d}x$$

$$\text{(i-y)} = \underbrace{\left[\overline{u} v'_y\right]_0^b}_{=0} - \int_0^b \overline{u}'_y v'_y \, \mathrm{d}y$$

$$\downarrow$$

$$(1) = \int_0^b \underbrace{\left(\int_0^a \overline{u}'_x v'_x \, \mathrm{d}x \right)}_{\text{(ii-x)}} \mathrm{d}y + \int_0^a \underbrace{\left(\int_0^b \overline{u}'_y v'_y \, \mathrm{d}y \right)}_{\text{(ii-y)}} \mathrm{d}x = (***)$$

$$\text{(ii-x)} = \underbrace{[\overline{u}'_x v'_x]_0^a}_{=0} - \int_0^a \overline{u}''_{xx} v \, \mathrm{d}x$$

$$\text{(ii-y)} = \underbrace{\left[\overline{u}'_y v'_y\right]_0^b}_{=0} - \int_0^b \overline{u}''_{yy} v \, \mathrm{d}y$$

$$\downarrow$$

$$(1) = -\int_0^b \int_0^a \overline{u}''_{xx} v \, \mathrm{d}x \, \mathrm{d}y - \int_0^a \int_0^b \overline{u}''_{yy} v \, \mathrm{d}y \, \mathrm{d}x = \int_0^a \int_0^b (-\overline{u}''_{xx} - \overline{u}''_{yy}) v \, \mathrm{d}x \, \mathrm{d}y =$$

$$= \boxed{(\mathcal{A}u \mid v)}$$

To show positive semidefinite, use the form described on (***) and $u = v$:

$$(***) = \int_0^a \int_0^b (\overline{u}'_x u'_x + \overline{u}'_y u'_y) \, \mathrm{d}x \, \mathrm{d}y = \int_0^a \int_0^b \left(|u'_x|^2 + |u'_y|^2 \right) \mathrm{d}x \, \mathrm{d}y \geq 0$$

To find the eigenvalues associated with the operator, use separation of variables

$$u(x, y) = X(x)Y(y)$$

$$\mathcal{A}u = \lambda u \to -X''Y - XY'' = \lambda XY \to -\frac{X''}{X} - \frac{Y''}{Y} = \lambda$$

$$\downarrow$$

$$\begin{cases} \frac{X''}{X} = -\gamma_x^2, & X(0) = X(a) = 0 \\ \frac{Y''}{Y} = -\gamma_y^2, & Y'(0) = Y'(b) = 0 \end{cases}$$

Starting with $X(x)$:

$$X'' = -\gamma_x^2 X \to X(x) = e^{i\gamma_x x} = A\cos(\gamma_x x) + B\sin(\gamma_x x)$$
$$X(0) = A = 0 \to A = 0$$
$$X(a) = B\sin(\gamma_x a) = 0 \to \gamma_x = \frac{\pi k_x}{a},\ k_x = 1, 2, 3...$$
$$\downarrow$$
$$X(x) = B\sin\left(\frac{\pi k_x x}{a}\right) \quad \square$$

Now $Y(y)$:

$$Y'' = -\gamma_y^2 Y \to Y(y) = e^{i\gamma_y y} = A\cos(\gamma_y y) + B\sin(\gamma_y y)$$
$$Y'(x) = -\gamma_y A\sin(\gamma_y y) + \gamma_y B\cos(\gamma_y y)$$
$$Y'(0) = \gamma_y B = 0 \to B = 0$$
$$Y'(b) = -\gamma_y A\sin(\gamma_y b) = 0 \to \gamma_y = \frac{\pi k_y}{b},\ k_y = 0, 1, 2...$$
$$\downarrow$$
$$Y(y) = A\cos\left(\frac{\pi k_y y}{b}\right) \quad \square$$

$\square$

H.8 Operators in Hilbert Spaces

Linear avbildning

$$\mathcal{A} : \mathcal{H}_1 \to \mathcal{H}_2$$

where $\mathcal{H}_1$ and $\mathcal{H}_2$ are two pre-Hilbert Spaces. Linearity:

$$\mathcal{A}(\lambda u + \mu v) = \lambda\mathcal{A}u + \mu\mathcal{A}v$$

is termed a **linear operator**.

Example H.19

Let I be a compact interval on $\mathbb{R}$, and let $K(x, y)$ be continuous for $x, y \in I$.

$$v(x) = \int_I K(x, y)u(y)\,\mathrm{d}y$$

This defines an **integral operator**

$$\mathcal{K} : L_2(I) \to L_2(I)$$

The function $K(x, y)$ is termed the operators **kernel**